

PROJECT IMPROVEMENTS & ARCHITECTURAL ROADMAP CHECKLIST

An Intelligent AI-Powered Restaurant Ordering System Using Augmented Reality

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Live Production URL: <https://smart-restaurant-2za8.vercel.app/> | Comprehensive 1-Page Master Action Checklist

1. SPATIAL AR/VR COMPUTING & 3D ENGINE ☐ WebRTC Synchronized Multi-User AR: Communal shared table 3D dish interaction at 60Hz ☐ Automated NeRF & 3D Gaussian Splatting: Instant 3D food asset pipeline from 45s chef video ☐ WebGPU Hardware Acceleration: High-fidelity fluid, sauce translucency & micro-textures at 120 FPS ☐ WebXR Ambient Light Estimation: Dynamic environmental illuminance & candlelit shadow matching ☐ Sub-Millimeter Metric Scale Clamping: Universal 1:1 real-world physical portion preservation ☐ Enhanced Cross-Platform Fallback: Native WebXR → Scene Viewer → Apple QuickLook USDZ	4. ENTERPRISE POS, ERP & BILLING INTEGRATION ☐ Global POS REST Webhooks: Certified connectors for Oracle Micros Symphony & Toast POS ☐ Regional Hospitality POS Integration: Direct terminal bridges for Petpooja and POSist ☐ Automated Real-Time 86'ing: Instant menu item graying upon kitchen ingredient depletion ☐ Live Wine Cellar Depletion API: Automated vintage bottle stock lock and inventory decrement ☐ Contactless Cloud Billing: Automated GST/VAT calculation, split-bill checks & UPI / Apple Pay ☐ Centralized Multi-Location Menu Sync: Chain-wide menu updates pushed in single click
2. ARTIFICIAL INTELLIGENCE & VISION QA ☐ On-Device Generative AI Voice Sommelier: Edge LLM wine pairings & cellar recommendations ☐ Overhead Computer Vision Kitchen QA: YOLOv8/ViT plate symmetry & garnish inspection at pass ☐ Dynamic Multi-Allergen Safety Engine: Real-time cross-contamination warnings & substitutions ☐ AI Dynamic Surge & Margin Pricing: Real-time yield management for desserts and cocktails ☐ Conversational Dish Storytelling: Interactive farm provenance, organic sourcing & chef notes ☐ Multilingual Voice Translation: Instant speech ordering in 12+ international languages	5. IOT HARDWARE & SUSTAINABILITY ENGINEERING ☐ Tabletop BLE Load-Cell Weight Coasters: Proactive beverage refill alerts at 15% threshold ☐ 100% Zero-Paper Circularity: Complete elimination of laminated plastic/paper printouts ☐ 28% Food Plate Waste Reduction: Mitigating order remorse via 1:1 true-scale pre-visualization ☐ Low-Power Mobile Edge Delivery: Draco mesh quantization saving device battery & mobile data ☐ Smart Table Turnover Sensors: Infrared occupancy sensors reporting real-time busing readiness ☐ ESG Sustainability Dashboard: Commercial carbon & food waste audit metrics for managers
3. KITCHEN OPERATIONS & KDS STATE ENGINE ☐ Distributed WebSocket & Redis Cluster: Zero-latency multi-kitchen terminal cloud synchronization ☐ Predictive Prep-Time Engine: AI ticket duration pacing & cook station load balancing ☐ Wearable Staff Haptic Notifications: Smartwatch runner alerts upon passing visual QA gate ☐ Station-Specific Acoustic Chimes: Frequency differentiation for rush, bar, and allergy orders ☐ Split-Station Ticket Routing: Automated order slicing across grill, pantry, bar, and pastry lines ☐ Kitchen Production Analytics: Real-time bottleneck detection & cook throughput metrics	6. STRATEGIC PIVOT TO LUXURY FINE DINING ☐ Target Market Realignment: Shifting from casual (AOV █500) to fine dining (AOV █1.8k–█5.5k+) ☐ High-End Survey Execution: Sampling N=150 luxury diners & N=25 Michelin/fine-dining chefs ☐ Multi-Course Tasting Menu Flights: Sequential 5-to-7 course AR journey through chef specials ☐ White-Tablecloth Luxury UI: Elegant minimalist typography, dark mode & discreet interactions ☐ VIP Guest Recognition: Private allergy records, preferred table seating & sommelier history ☐ Empirical Validation & Publication: Finalizing IEEE conference research paper submission
Phase 1 (Live Now): Zero-Install WebXR 1:1 Metric Clamping Live KDS Acoustic Chimes Vercel CDN	Phase 2 (Horizon 1): Luxury Fine Dining Pivot AI Voice Sommelier Reset Survey Execution
Phase 3 (Horizon 2): WebRTC Multi-User Table AR Automated NeRF & Gaussian Splatting Pipeline	Phase 4 (Horizon 3): Kitchen Plating Vision QA Gate Oracle Micros & Toast POS ERP Webhooks